

2	(a)	$\Delta KE = KE_{\text{final}} - KE_{\text{initial}}$ $= \frac{1}{2}mv^2 - 0$ $= \frac{1}{2}(60)(55)^2 \quad [\text{M1:reading and substitution}]$ $= 90750$ $\approx 91000 \text{ J} \quad [\text{A1}]$																														
	(b)	finding height dropped using area under graph High precision answer for tutor reference (using 30 trapeziums): height dropped = 1422.5 m Marking scheme 1 using three trapeziums (minimum) $\text{height dropped} = \frac{1}{2}(0 + 35)(5 - 0) + \frac{1}{2}(55 + 35)(11 - 5) + (55)(30 - 11) \quad [\text{M1}]$ $= 87.5 + 270 + 1045$ $= 1402.5$ $\approx 1400 \text{ m} \quad [\text{A1}]$ Marking scheme 2 counting squares (11 squares) $\text{height dropped} = 11 \times (25 \times 5) \quad [\text{B1}]$ $= 1375$ $\approx 1400 \text{ m}$ <table border="1"><thead><tr><th colspan="2">read-offs</th></tr><tr><th>t</th><th>v</th></tr></thead><tbody><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>10</td></tr><tr><td>2</td><td>17.5</td></tr><tr><td>3</td><td>22.5</td></tr><tr><td>4</td><td>30</td></tr><tr><td>5</td><td>35</td></tr><tr><td>6</td><td>40</td></tr><tr><td>7</td><td>45</td></tr><tr><td>8</td><td>47.5</td></tr><tr><td>9</td><td>50</td></tr><tr><td>10</td><td>52.5</td></tr><tr><td>11</td><td>55</td></tr><tr><td>12</td><td>55</td></tr></tbody></table> Accepted range: Answers (before rounding) that rounds off to 1400 m (2 s.f.) Minus 1 mark if answer not expressed to 2 s.f. as instructed.	read-offs		t	v	0	0	1	10	2	17.5	3	22.5	4	30	5	35	6	40	7	45	8	47.5	9	50	10	52.5	11	55	12	55
read-offs																																
t	v																															
0	0																															
1	10																															
2	17.5																															
3	22.5																															
4	30																															
5	35																															
6	40																															
7	45																															
8	47.5																															
9	50																															
10	52.5																															
11	55																															
12	55																															



	(c)	$GPE_{loss} = mg\Delta h$ $= (60)(9.81)(1400)$ $= 824040$ $\approx 820000 \text{ J} \quad [\text{B1}]$
	(d)	work done by resistive force + loss in GPE = gain in KE $\text{work done by resistive force} = 91000 - 820000 \quad [\text{C1}]$ $= -729000$ magnitude of work done by resistive force $\approx 730000 \text{ J} \quad [\text{A1}]$

3	(a)		The electric field strength at a point is the electric force per unit positive
---	-----	--	--